

Rohan

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EDUCATION

PES University

Bachelor of Technology in Computer Science & Engineering

Bengaluru, India

2023 – 2027

- CGPA: 7.86/10
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Compiler Design, Cloud Computing, Generative AI, Object-Oriented Analysis and Design, Software Engineering.

EXPERIENCE

UI/UX Web Developer Intern

March 2024 – April 2025

InstiBuzz

Remote/IIT-Madras

- Engineered responsive UI components using **Figma** and **HTML/CSS**, significantly improving mobile accessibility and user engagement metrics.
- Implemented clean, minimalist UI patterns that streamlined user workflows, focusing on intuitive navigation and consistent design language to improve the overall usability of client-facing features.
- Collaborated with a cross-functional team of engineers using **Git** for version control within an Agile development workflow.

PROJECTS

Policy-Aware Code Compliance Auditor | [GitHub Repo](#) | *PyTorch, CodeBERT, LLaMA 3.1, HuggingFace, FAISS* Apr 2026

- Developed a hybrid AI system to automatically audit large-scale Python codebases against business policies, achieving a **93% compliance detection rate** in live testing.
- Performed end-to-end data preprocessing on HuggingFace datasets using **Pandas** and **NumPy** (missing labels, deduplication, tokenization) before fine-tuning a **CodeBERT** classifier with **LoRA (PEFT)**.
- Designed a hybrid routing engine using the classifier as a fast gatekeeper, applying **batching and inference optimization** to **reduce expensive LLM (Kimi K2) reasoning calls by 60%**.
- Integrated **Gemini Embeddings** and a **FAISS** vector database for semantically relevant policy retrieval (**RAG**), cutting inference costs by 75%; conducted controlled benchmarking and **documented** error analysis to iteratively improve model performance.

Smart Glove for Assistive Healthcare Communication | [GitHub Repo](#) | *C++, Arduino, I2C, Bluetooth* Jan 2025

- Developed a wearable **assistive-technology** device to improve **healthcare accessibility** for the hearing-impaired, using an Arduino microcontroller with four flex sensors and an MPU6050 module via I2C for real-time hand orientation capture.
- Engineered a pattern-matching system to process sensor data streams, mapping hand movements to predefined sign language gestures and converting them into text – enabling non-verbal communication in clinical settings.
- Implemented an HC-05 Bluetooth module for wireless transmission of translated text to external devices such as smartphones or hospital displays.

Airline Reservation System (ARS) | [GitHub Repo](#) | *React, Spring Boot, Java 17, Firebase* April 2026

- Engineered a responsive SPA using React and Vite with a secure proxy routing all HTTP requests to a Spring Boot backend; **documented** system architecture and design patterns for team reference.
- Architected the persistence layer using Firebase Firestore (NoSQL), applying the GRASP Pure Fabrication principle to isolate data access from business logic.
- Built an interactive seat mapping interface reflecting real-time availability and synchronizing concurrent booking states between client and database.

Real-Time Streaming Platform | [GitHub Repo](#) | *MERN Stack, WebSockets* July 2025

- Architected a live streaming application achieving **sub-100ms latency** using **WebSockets** and **Node.js**.
- Implemented secure room-based routing to strictly prevent unauthorized streaming between private sessions.
- Developed a responsive frontend in **React.js** to handle multiple concurrent data streams of text and video.

LEADERSHIP & EXTRACURRICULAR

Vice Chair & Secretary

August 2025 – Present

IEEE PESU Student Branch

Bengaluru, India

- Led the technical direction of the student branch, designing an advanced AI and cybersecurity curriculum for workshops.
- Designed and implemented custom cybersecurity challenges including Web Exploitation and Cryptography for the yearly CTF hackathon, overseeing competition infrastructure for **200+** participants.
- Reviewed code and offered technical mentorship to junior members, helping them become more proficient in ML and software development.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript (ES6+), SQL, Java

AI / ML: NumPy, Pandas, Scikit-learn, PyTorch, HuggingFace Transformers, LoRA/PEFT, CodeBERT, LLaMA 3.1, Gemini Embeddings, FAISS, RAG, NLP Pipelines, Prompt Engineering, LLM Fine-Tuning, Transformer Architectures

Web & Tools: React.js, Node.js, Express.js, Spring Boot, HTML5, CSS3, REST APIs, WebSockets, Git, Docker, Postman, MongoDB, Firebase Firestore